

ObSchema: From Local Observations to Global Entity Attribute Schemas in Scientific Literature

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Abstract

Scientific literature reports fine-grained entity properties, but such knowledge is scattered across papers, expressed with heterogeneous names, and scoped by experimental conditions. Closed-schema extraction requires predefined attribute vocabularies that are hard to enumerate for property-rich scientific entities, while open-schema extraction often yields duplicate and condition-specific attribute labels. We introduce entity-centric attribute schema induction, which constructs a hierarchical type-level attribute schema from a scientific corpus. We propose ObSchema, an observation-to-schema framework that extracts local entity-property observations, aggregates corpus-level evidence to identify recurring attributes, and refines them through value-aware merging and adaptive grouping. On two physical science domains, ObSchema consistently outperforms strong LLM-based schema-induction baselines across different backbone models, producing more compact and coherent schemas.

1 Introduction

Scientific literature contains fine-grained entity-property knowledge essential for tasks such as scientific retrieval (Kang et al., 2024; Kweon et al., 2026), reasoning (Bran et al., 2024; Parekh et al., 2026), and discovery (Su et al., 2025; Stewart et al., 2026). However, this knowledge is rarely structured: it is scattered (Dagdelen et al., 2024), expressed with heterogeneous names (Zhang et al., 2024), and tied to the conditions under which it was observed (Panapitiya et al., 2021).

A common way to mine such knowledge is closed-schema extraction, where the model extracts values for a predefined attribute vocabulary (Zheng et al., 2018; Yang et al., 2022; Wu et al., 2024). This works for small, well-specified schemas, but scientific entity types are studied along too many complementary dimensions to enumerate in advance (Hong et al., 2021; Durmaz et al.,

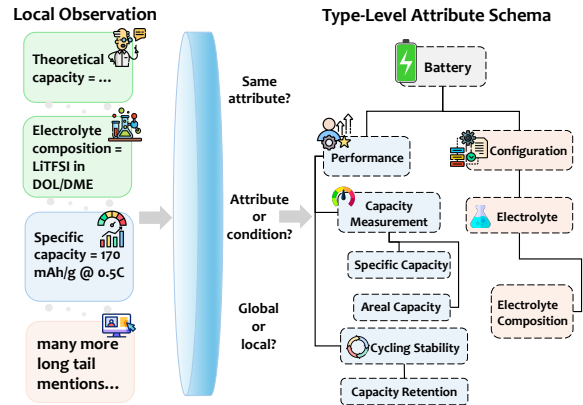


Figure 1: Motivation of entity-centric attribute schema induction. Local scientific observations mix attributes, values, units, and conditions, while the desired schema should capture recurring type-level attributes and organize them into coherent scientific dimensions.

2024)—battery cells alone span performance, cycling stability, composition, configuration, and fabrication (Lee et al., 2025). Narrow vocabularies then miss salient attributes, but broad ones accumulate rare attributes irrelevant to most papers.

Open-schema methods relax this by generating attribute names directly from text (Lu et al., 2023; Jiao et al., 2023), but the resulting attribute set is unstable: the same property surfaces under different names, and measurement conditions get absorbed into composite labels (Zhang and Soh, 2024). For example, a single battery paragraph may yield *capacity*, *specific capacity*, *discharge capacity at 0.5C*, and *capacity retention after 300 cycles*, which are related-but-distinct properties with property-condition compounds, poorly suited for organizing knowledge.

We therefore ask a different question: rather than fixing the vocabulary in advance or accepting an unstable one, can we induce an attribute schema for a target entity type directly from the corpus itself? Such a schema should not attempt to enumerate every possible property in the domain. Instead,

it should capture the recurring attributes that are salient in the given corpus, normalize heterogeneous mentions, and organize them into meaningful scientific dimensions. We refer to this task as *entity-centric attribute schema induction*.

Prior work on schema induction from text (Jiao et al., 2022; Bai et al., 2026; Kargupta et al., 2025) organizes concept labels at the granularity of topics, entity types, or event roles, where topical or semantic similarity suffices to merge or split nodes. On the other hand, entity-centric attribute schemas are structurally different: each node is a fine-grained property whose identity depends on values, units, and conditions—*capacity* (150 mAh) and *specific capacity* (170 mAh/g) look nearly synonymous as attribute names, yet the units alone reveal them as distinct quantities, one absolute and one mass-normalized. Inducing such a schema requires evidence-aware control over fine-grained contextual signals.

To enable such control, we propose **ObSchema**, an observation-to-schema framework that aggregates and organizes entity properties from local documents to incrementally construct a global hierarchical entity type attribute schema. Specifically, ObSchema processes documents in batches through three stages. ObSchema first extracts each entity-property report from a paragraph as an *observation*, an n -tuple that bundles entity, attribute name and definition, value, conditions, and category. It then aggregates observations into candidate attributes and promotes only those with sufficient corpus-level evidence into the schema. Finally, we refine the schema via conservative duplicate merging and adaptive grouping, yielding a compact hierarchical attribute schema.

We evaluate ObSchema using different backbone large language models (LLMs) on two scientific domains: Battery Science and Geoscience. The results demonstrate that ObSchema consistently outperforms strong schema-induction baselines across both domains and backbones. These results demonstrate the effectiveness and generalizability of our framework for scientific attribute schema induction.

Our contributions are summarized as follows:

- We study **entity-centric attribute schema induction**, the problem of constructing hierarchical attribute schemas for entity types from a scientific corpus.
- We propose **ObSchema**, a framework that builds a hierarchical entity type attribute

schema through three stages: observation extraction, evidence-based attribute organization, and schema refinement.

- We evaluate ObSchema on two property-rich physical science domains and across multiple backbone LLMs, showing consistent improvements over strong schema-induction baselines.

2 Related Work

2.1 Entity-Centric Information Extraction

Entity-centric information extraction has traditionally been formulated as a closed-schema problem: given a predefined set of attributes, the system extracts their corresponding values from text (Yang et al., 2022; Zheng et al., 2018). Such methods are effective when the target schema is available, because the schema constrains the output space to known entity types, attributes, and data formats, allowing the model to focus on evidence identification and field filling (Zhong et al., 2023; Bai et al., 2024; Wu et al., 2024; Dagdelen et al., 2024). However, in scientific domains, the relevant attributes of an entity type are often broad, evolving, and only partially known before reading the corpus, which shifts much of the knowledge organization burden to manual schema design (Tshitoyan et al., 2019; Pan et al., 2025; Venugopal and Olivetti, 2024).

Open-schema extraction relaxes this assumption by generating attribute or relation names from text (Lu et al., 2023). For example, ODIE (Jiao et al., 2023) produces flexible entity-property tables, EDC (Zhang and Soh, 2024) induces and canonicalizes relation schemas from open IE triples, and ZOES (Xu et al., 2026) discovers open-schema entity structures through iterative entity-attribute-value enrichment. Our work is complementary to these methods: open-schema extractors provide rich local observations from documents, while schema induction aggregates such observations across a corpus, unifies recurring attributes and organizes them into a reusable type-level attribute ontology.

2.2 Schema and Taxonomy Induction

Schema and taxonomy induction aims to organize open-ended concepts, relations, or roles into reusable structures from text. Prior work has studied open vocabulary event-role schemas (Jiao et al., 2022; Shi et al., 2026), taxonomy construction (Shen et al., 2020; Zeng et al., 2024; Sun et al., 2026), and dynamic schema induction for

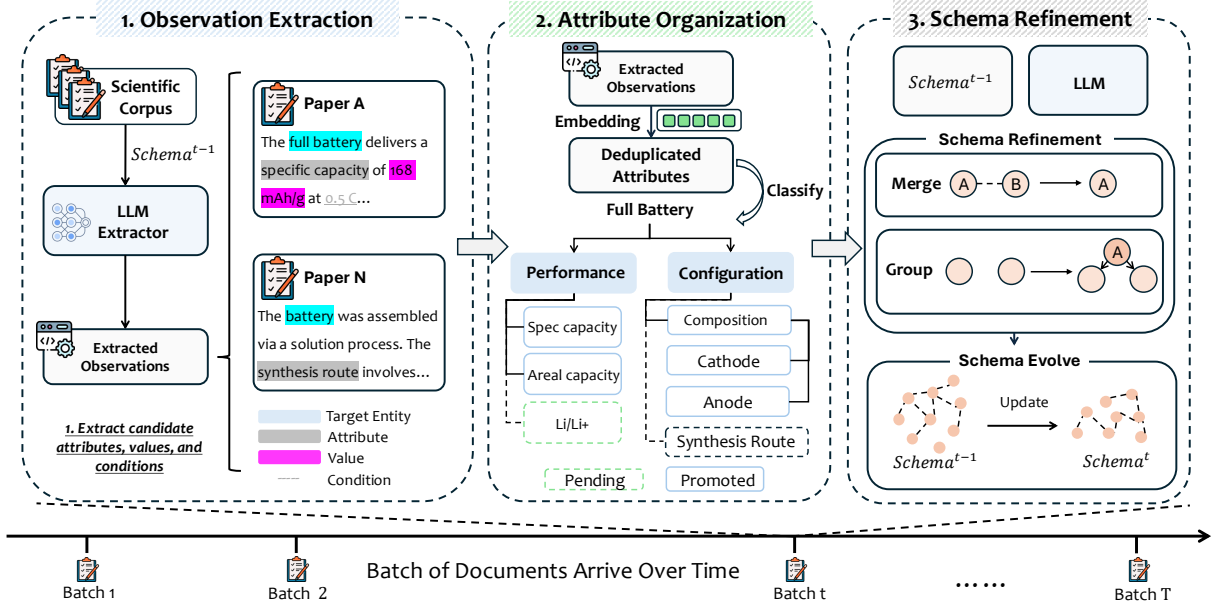


Figure 2: Overview of ObSchema. Given batches of scientific documents about a target entity type, ObSchema incrementally induces a hierarchical attribute schema through three stages. **Observation Extraction** extracts n -ary observations binding the entity, candidate attribute, value, category, and measurement conditions, guided by the current schema S_{t-1} . **Attribute Organization** consolidates near-identical raw attributes, classifies them into the schema, and promotes them into the active schema only after accumulating sufficient corpus-level evidence. **Schema Refinement** then applies conservative duplicate merging and adaptive grouping to update S_{t-1} into S_t .

knowledge graph construction (Bai et al., 2026). These methods are related to our goal because they move beyond extracting isolated facts and seek corpus-level organization. However, most existing schema-induction settings organize topical concepts (Lee et al., 2022; Jiang et al., 2022; Kargupta et al., 2025), entity types, event roles, or relation labels (Shen et al., 2020), where semantic similarity is often sufficient to propose parent–child relations.

Entity-centric attribute schemas in scientific domains pose a different knowledge-intensive challenge (Cheung et al., 2024). An attribute is not merely a concept label, but a measurable property of a target entity type. Whether two attributes should be merged or separated often depends on their values, units, measurement contexts, and scientific definitions (Panapitiya et al., 2021; Gupta et al., 2024): for example, *capacity*, *specific capacity*, and *capacity retention* are lexically related but denote different properties. Our work therefore treats local entity-property mentions as observation evidence and uses LLMs as knowledge-aware aggregators to interpret and organize them. Rather than constructing hierarchies from similarity alone, we combine the LLM’s parametric knowledge of scientific concepts with corpus-level evidence from values, units, definitions, and measurement con-

texts, enabling evidence-grounded attribute promotion, conservative merging, and adaptive grouping.

3 Method

Figure 2 illustrates the overall framework. ObSchema processes the corpus in batches, allowing the schema to update as new documents arrive through observation extraction, attribute organization and promotion, as well as schema refinement.

3.1 Problem Formulation

We study entity-centric schema induction from scientific documents. Given an unlabeled corpus of paragraphs $D = \{d_1, \dots, d_n\}$ describing a target entity type τ , and a small set of human-specified coarse categories $\mathcal{C} = \{c_1, \dots, c_k\}$, our goal is to induce a hierarchical schema $\mathcal{S}$ that captures the scientific attributes studied for entities of type τ . We represent $\mathcal{S}$ as a tree-structured attribute ontology, where each node is a coarse category, an induced sub-category, or a fine-grained attribute, and each non-root node has exactly one parent.

As shown in Figure 1, for the entity type *battery cell*, the schema may organize attributes under coarse categories such as *Cell Performance* and *Cell Configuration*, with attributes such as *specific*

capacity placed along single paths in the tree. The goal is not to include every property mentioned in the corpus: attributes that only apply to a narrow subtype may be locally meaningful, but should not dominate the general schema. We therefore seek a schema that captures recurring, type-level attributes of τ .

3.2 Observation Extraction

The first stage converts paragraph-level scientific descriptions into structured observation evidence. In scientific text, entity attributes are typically reported together with experimental or measurement conditions, so naive extraction yields overly specific composite attributes such as *coulombic_efficiency_at_high_temperature* or *capacity_at_0.5C*, when the desired schema attributes are *coulombic efficiency* and *capacity*.

Rather than (subject, relation, object) triples (Banko et al., 2007; Pai et al., 2024), we extract paragraph-level facts as n -ary observation tuples that explicitly separate the candidate attribute from its supporting conditions. Specifically, for each paragraph in a batch B_t , we prompt an LLM with the current schema $\mathcal{S}_{t-1}$ as soft guidance to extract observations of the form

$$o = (e, a_{\text{raw}}, \delta, v, c, Q),$$

where e is the entity mention, a_{raw} is the raw attribute name, δ is a one-sentence natural-language definition of a_{raw} generated in the paragraph context, v is the extracted value, $c \in \mathcal{C}$ is the coarse category, and $Q = \{(q_i, w_i)\}$ is a set of condition type-value pairs scoping v , such as temperature, cycle number, or C-rate.

3.3 Attribute Organization and Promotion

The second stage decides where each raw observation attribute belongs in the schema and when it has enough evidence to be committed. We separate *semantic placement*—the parent node a candidate belongs under—from *schema commitment*—whether the candidate has accumulated enough corpus-level evidence to become an active node. Duplicate verification is deferred to refinement (§3.4), where promoted nodes carry richer multi-batch evidence.

Hierarchical classification. Within each batch, we first apply a lightweight surface-level pre-merge: raw attributes whose (value, description) embeddings exceed a 0.95 cosine threshold, indicating almost the same attributes, are collapsed into a single

candidate, reducing trivial surface noise before classification. Following prior work on LLM-assisted hierarchical classification (Zhang et al., 2025), we then route each candidate through a two-stage procedure: a *category gate* that routes the candidate to one of the coarse categories (or rejects it as a non-property), followed by *top-down descent* within that category to identify the best-matching parent in the existing hierarchy.

Evidence-based promotion. A classified candidate is not immediately incorporated into the active schema. Instead, it is held in a pending pool and accumulates corpus-level evidence across batches, so that LLM-side semantic placement is decoupled from a deterministic schema-commitment decision.

The evidence score $s(a) \in [0, 1]$ is the geometric mean of two complementary signals: a *popularity* term that grows with how broadly a recurs across both source documents and entity mentions, and a *category-relative share* term that compares a ’s entity support against the leading leaf attribute in its category. The geometric mean penalizes any candidate that lacks either signal, so an attribute enters the schema only when it is corpus-wide *and* locally prominent. A pending attribute is promoted once $s(a) \geq \theta$; promoted attributes become visible to subsequent extraction prompts, while pending ones remain in the pool and are rescored as new evidence arrives. The full functional forms and the threshold value $\theta = 0.15$ are given in Appendix B.2.

3.4 Schema Refinement

The promoted schema may still contain near-duplicate leaves and overly flat branches: the surface-level pre-merge during batch processing (§3.3) only collapses near-identical embeddings, leaving both lexically distant synonyms and cross-batch duplicates unresolved. ObSchema therefore periodically refines the active schema.

Schema-level duplicate merging. We retrieve candidate duplicates among promoted leaves within each coarse category via embedding similarity, then ask an LLM to verify whether they denote the same underlying scientific quantity. Verification is *value-aware*: representative values, units, definitions, and supporting evidence are passed to the LLM, because surface names alone can collapse related-but-distinct quantities in scientific domains (e.g., absolute vs. mass-normalized capacity). Confirmed merges pool evidence, surface forms, children, and condition statistics into the kept node.

The merge is intentionally conservative. Lexically close attributes may denote distinct measurements. For example, *specific capacity* and *areal capacity* differ in normalization basis, and attributes differing by subject, measurement direction, temporal scope, or experimental definition are likewise kept separate. This prevents refinement from improving compactness at the cost of losing scientifically meaningful distinctions.

Adaptive grouping. Even after merging, a coarse category may contain many fine-grained leaves with no natural parent-child relations among themselves, leaving the hierarchy flat and hard to navigate. Surface-level clustering is unreliable here: *specific capacity* and *capacity retention* share lexical roots but describe distinct dimensions (mass-normalized storage vs. cycling stability).

ObSchema therefore introduces intermediate parents through an evidence-grounded, three-step procedure. (i) *Aspect induction*: the LLM proposes a compact set of organizing aspects from the leaves under each coarse category, each describing a scientific dimension rather than a literal schema node. (ii) *Group formation*: we match each leaf to its best-supporting aspects using its name, definition, representative values, units, and supporting observations, yielding candidate parent-node proposals. (iii) *Consolidation*: the LLM merges overlapping proposals and instantiates an optional parent only when the grouped leaves share a coherent dimension. Otherwise the proposal is rejected and the affected leaves remain under their original parent. This admits intermediate structure where the evidence supports it without forcing a fixed depth. All prompts are in Appendix G.

4 Experiments

In this section, we first describe the dataset construction and experimental setup. We then evaluate ObSchema through main experiments, followed by analysis studies that reveal its characteristics.

4.1 Dataset Construction

We construct a paragraph-level scientific corpus from two domains, battery science and geoscience. We choose these domains because entities in both areas, including battery cells and geoscientific samples, are characterized along many complementary dimensions (e.g., composition, performance, fabrication/formation, and measurement context), yielding broad attribute spaces that are challenging

for entity-centric schema induction. For each domain, we collect full-length scientific articles from domain-relevant journals, segment them into paragraphs, and retain only entity-centric passages with extractable properties, measurements, or configurations. As summarized in Table 1, the resulting corpus comprises 1,305 paragraphs from 303 articles across 37 venues spanning 2007–2025, ensuring that the induced schema is not biased toward a single research group. Full filtering criteria and paragraph length statistics are provided in Appendix C.

4.2 Experiment Setup

Base models. We evaluate ObSchema with two backbone LLMs spanning different architectures. The primary base model is Qwen3-30B-A3B (Yang et al., 2025), a mixture-of-experts (MoE) model with 30B total parameters but only ~ 3 B activated per token. To verify that our method generalizes beyond a single architecture, we additionally report results with Qwen3-14B (Yang et al., 2025), a dense model at a comparable scale.

Implementation. We process each domain in 10 incremental batches and use Qwen3-Embedding-8B (Yang et al., 2025) for embedding similarity calculation. A pending attribute is promoted to the schema when its evidence score exceeds 0.15. More details can be found at Appendix B.2. The human-specified coarse category set $\mathcal{C}$ is $\{\textit{cell performance}, \textit{cell configuration}\}$ for battery science and $\{\textit{petrophysical properties}, \textit{composition}\}$ for geoscience. All steps share the same base LLM. Experiments run on $2 \times$ NVIDIA GH200 GPUs. All LLMs are served with vLLM with 32-way parallelism.

Metrics. Following previous studies (Kargupta et al., 2025), we evaluate the induced schema along four complementary axes, each scored by an LLM judge with GPT-5-mini (Singh et al., 2026) as the backbone model. The detailed rubrics are provided in Appendix E.

- **Attribute Validity (AV)** measures whether each leaf node denotes a real, measurable property of the target entity type τ , as opposed to a procedural step or out-of-scope concept.
- **Path Granularity (PG)** measures whether every root-to-leaf path forms a strict refinement chain, with each child being a more specific sub-property of its parent.

Table 1: Statistics of our scientific article corpus across two domains (battery science and geoscience), reporting article and paragraph counts, average paragraphs per article (Para/Art), average paragraph length in words, publication year range, and number of source venues.

Domain	#Articles	#Paragraphs	Para/Art	Avg words/Para	Year Range	#Venues
Battery Science	185	927	5.0	240	2015–2025	13
Geoscience	118	378	3.2	267	2007–2025	24
Total	303	1,305	4.3	247	2007–2025	37

Domain	Method	Attr. Validity	Path Gran.	Sibling Coh.	Uniqueness	Avg.
Base model: Qwen3-30B-A3B						
Battery	Chain-of-Layer	0.860	0.711	0.667	0.383	0.655
	TaxoAdapt	0.951	0.717	0.626	0.559	0.713
	Iterative LLM Induction	0.935	0.665	0.756	0.733	0.772
	ObSchema (ours)	0.964	0.807	0.816	0.830	0.854
Geoscience	Chain-of-Layer	0.950	0.703	0.708	0.594	0.739
	TaxoAdapt	0.964	0.800	0.630	0.690	0.771
	Iterative LLM Induction	0.891	0.589	0.556	0.724	0.690
	ObSchema (ours)	0.991	0.798	0.818	0.807	0.853
Base model: Qwen3-14B						
Battery	Chain-of-Layer	0.817	0.683	0.538	0.447	0.621
	TaxoAdapt	0.976	0.808	0.548	0.651	0.746
	Iterative LLM Induction	0.912	0.537	0.444	0.658	0.638
	ObSchema (ours)	1.000	0.935	0.667	0.704	0.827
Geoscience	Chain-of-Layer	0.920	0.802	0.606	0.564	0.723
	TaxoAdapt	0.919	0.787	0.539	0.679	0.731
	Iterative LLM Induction	0.754	0.521	0.333	0.771	0.595
	ObSchema (ours)	1.000	0.864	0.800	0.911	0.894

Table 2: Schema induction quality on the Battery and Geoscience corpora under two base LLMs. Methods are evaluated against four metrics, including *Attribute Validity*, *Path Granularity*, *Sibling Coherence*, and *Uniqueness*, with **Avg.** representing the mean of other metrics. Best results are in **bold**.

- **Sibling Coherence (SC)** measures whether the children of each internal node are placed at a consistent level of abstraction.
- **Uniqueness (Uniq.)** measures the absence of semantically duplicate nodes in the schema.

Baselines. We compare with three LLM-based schema induction baselines: Chain-of-Layer (Zeng et al., 2024), which induces a hierarchy in a top-down, layer-by-layer manner; TaxoAdapt (Kargupta et al., 2025), which adapts a multidimensional schema to a target corpus; and Iterative LLM Induction, which streams paragraphs to an LLM that maintains a running schema, relying on the LLM’s inherent structuring capability (Yang et al., 2026). Details are provided in Appendix D.

4.3 Main Experiments

ObSchema outperforms all baselines on average across both domains and backbone LLMs (Table 2). With Qwen3-30B-A3B, it achieves average scores of 0.854 on Battery and 0.853 on Geoscience. The advantage carries over to the smaller dense Qwen3-14B, where ObSchema reaches 0.827 on Battery and 0.894 on Geoscience.

We highlight three findings from Table 2. First, the largest gains over the strongest baseline come from uniqueness (+9.7 on Battery, +11.7 on Geoscience with Qwen3-30B-A3B), the metric with the highest judge–human agreement in our reliability study (85%, Appendix E). This indicates that ObSchema’s value-aware merging and evidence-gated promotion reduce duplicate attributes more effectively than baselines that organize attributes by lexical or semantic similarity alone. Second, gains on sibling coherence follow the same direction,

Variant	AV	PG	SC	Uniq.	Avg.
Battery					
ObSchema	0.961	0.833	0.788	0.833	0.853
<i>w/o Classification</i>	0.929	0.718	0.619	0.772	0.760
<i>w/o Refinement</i>	0.951	0.817	0.595	0.619	0.745
Geoscience					
ObSchema	0.992	0.901	0.785	0.807	0.871
<i>w/o Classification</i>	0.968	0.748	0.559	0.778	0.763
<i>w/o Refinement</i>	0.986	0.928	0.528	0.725	0.791

Table 3: Ablation of ObSchema. AV: Attribute Validity, PG: Path Granularity, SC: Sibling Coherence, Uniq.: Uniqueness. Best per column in **bold**.

though we interpret this metric more cautiously given its lower judge agreement (70%). Third, on Qwen3-14B ObSchema reaches perfect attribute validity on both domains, suggesting that evidence-based promotion filters overly specific or out-of-scope candidates before they enter the schema—an effect that becomes especially pronounced when the backbone is smaller and more prone to noisy proposals.

The pattern across metrics is consistent: base-lines often produce attribute candidates that are individually plausible but fail to organize them into compact, non-redundant hierarchies. By separating candidate extraction, hierarchical placement, evidence-based promotion, and schema refinement, ObSchema addresses this organizational gap rather than the candidate-identification step alone.

4.4 Further Analysis

Ablation studies. Both hierarchical classification and schema refinement contribute substantially to ObSchema, as shown by the ablation in Table 3 on Qwen3-30B-A3B with a single run. Removing hierarchical classification drops the average score from 0.853 to 0.760 on Battery and from 0.871 to 0.763 on Geoscience, confirming that category routing and top-down descent are needed to filter invalid candidates and place attributes at appropriate positions in the schema. Removing schema refinement mainly hurts structure-sensitive metrics showing that schema-level duplicate merging and adaptive grouping are necessary to organize flat attributes into coherent groups.

Backbone analysis. ObSchema remains robust across LLM backbones under the same pipeline and hyperparameters. As shown in Table 4, both Qwen3-30B-A3B and Mistral-Small-24B-Instruct-2501 (Mistral

Backbone	AV	PG	SC	Uniq.	Avg.
Battery					
	0.964	0.807	0.816	0.830	0.854
	0.976	0.827	0.771	0.882	0.864
	0.885	0.789	0.750	0.628	0.763
Geoscience					
	0.991	0.798	0.818	0.807	0.854
	1.000	0.906	0.776	0.862	0.886
	0.972	0.814	0.759	0.783	0.832

Table 4: Backbone robustness of ObSchema under fixed pipeline and hyperparameters. Icons denote  Qwen3-30B-A3B (Yang et al., 2025),  Mistral-Small-24B-Instruct-2501, and  GPT-5-nano (Singh et al., 2026). Avg. is the mean of AV, PG, SC, and Uniq.; best per column within each domain is in **bold**.

AI, 2025) exceed an average score of 0.85 on Battery and Geoscience, with Mistral best on both domains, likely reflecting its strength on knowledge-intensive decisions such as distinguishing near-duplicate attributes and forming coherent groups. GPT-5-nano trails, particularly on Battery uniqueness, but still attains reasonable averages of 0.763 and 0.832. The design provides a stable control that generalizes across model families, while stronger backbones further sharpen fine-grained structural decisions.

Schema performance across batches. ObSchema’s design produces stable schemas as documents accumulate (Figure 3). Attribute validity stays consistently high across batches, indicating that observation extraction and evidence-based promotion maintain a stable notion of valid attributes. Sibling coherence and uniqueness fluctuate more in early batches when evidence is limited, but stabilize as additional batches arrive, while the total number of attributes remains controlled. This confirms that ObSchema can grow the schema without being dominated by early noise.

Efficiency studies. ObSchema’s batch-processing design enables parallel inference, making it the fastest method on both domains: 14.00 minutes on Battery and 8.67 minutes on Geoscience (Table 5). The gap is largest against Iterative LLM Induction, which processes paragraphs sequentially and updates the global schema after each, preventing parallelization across paragraphs. ObSchema instead runs observation extraction and hierarchical classification across

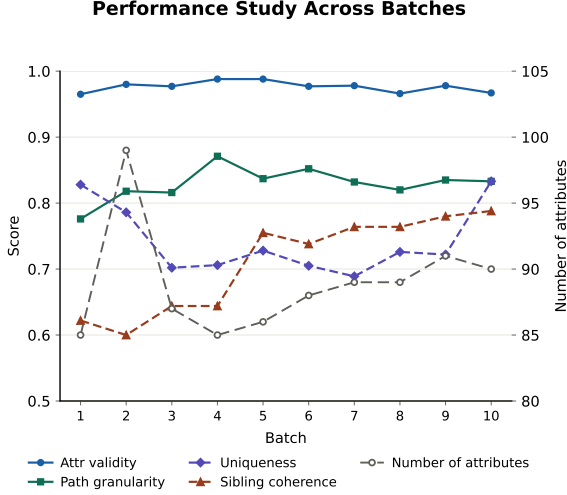


Figure 3: Performance of ObSchema across 10 incremental batches. The left axis reports four schema-quality scores while the right axis tracks the number of extracted attributes.

Domain	Method	Runtime (min)	Tokens (M)
Battery	Chain-of-Layer	14.73	8.30
	TaxoAdapt	21.55	31.92
	Iterative LLM	60.42	7.20
	ObSchema	14.00	13.22
Geoscience	Chain-of-Layer	9.44	6.43
	TaxoAdapt	13.78	11.34
	Iterative LLM	23.71	3.92
	ObSchema	8.67	5.33

Table 5: End-to-end runtime and token consumption for entity extraction and schema induction on Battery and Geoscience. All methods use Qwen3-30B-A3B on the same hardware with 32-way parallel inference; tokens are reported in millions.

an entire batch in parallel, and invokes schema refinement only at batch boundaries.

Extraction quality with induced schemas. Beyond schema structure itself, ObSchema’s induced schema also yields modest but consistent improvements on downstream entity-property extraction (Figure 4). We sample 100 paragraphs from the battery domain, use each induced schema as the closed schema for extraction, and pairwise compare ObSchema’s extractions against each baseline’s on coverage and faithfulness, judged by GPT-5-mini with A/B order randomized. ObSchema achieves win rates above the 0.5 tie line on both metrics and against all three baselines, with margins ranging from 0.517 to 0.600. While individual margins are small, the direction is consistent across baselines and metrics, suggesting that schemas with fewer

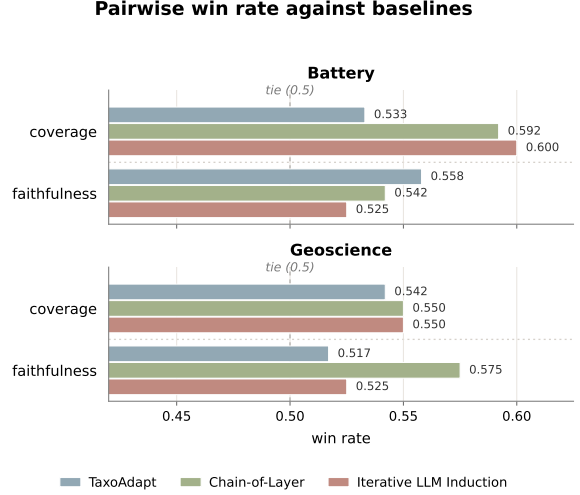


Figure 4: Pairwise win rate against baseline schemas on closed-schema extraction, judged by GPT-5-mini on coverage and faithfulness (A/B randomized).

Coverage (vs. 60-attribute expert reference)

Captured in final schema	83.3%
Missed (pending due to insufficient evidence)	16.7%

Precision (over 82 promoted nodes)

Core or valid niche attribute	98.8%
Redundant	1.2%
Invalid	0.0%

Table 6: Reference-based validation on the battery schema. ObSchema’s errors are coverage errors rather than precision errors. Per-category breakdowns, the node-level audit, and an attribute-level placement comparison against the baselines are given in Appendix F.3.

redundant or overly specific attributes translate into somewhat more complete and faithful extractions when used as a closed vocabulary.

Reference-based validation. To complement the LLM-judge metrics, we evaluate the battery schema against a 60-attribute reference list curated by a domain expert with graduate-level training in battery science, constructed independently of any ObSchema output (Appendix F.3). Table 6 reports the two sides of the comparison. ObSchema captures 83.3% of the reference attributes, while a node-level audit finds 98.8% of promoted nodes to be either core or scientifically valid niche attributes, with zero invalid promotions and a single residual duplicate. Together these numbers describe a deliberate asymmetry: ObSchema’s failures are almost entirely coverage errors. Rare-but-canonical attributes such as *voltage plateau* are extracted but

remain in the pending pool when their corpus-level evidence is insufficient.

5 Conclusion

We presented ObSchema, a framework for inducing entity-centric attribute schemas from scientific literature. By converting local entity-property mentions into structured observations, aggregating corpus-level evidence, and refining promoted attributes into coherent hierarchies, ObSchema bridges the gap between open-schema extraction and global scientific schema construction. Experiments on Battery Science and Geoscience show consistent gains over strong schema-induction baselines across multiple backbone LLMs.

Limitations

We acknowledge several limitations.

Domain coverage. Our evaluation covers two scientific domains, which are both within the physical sciences. Whether ObSchema generalizes to areas with substantially different reporting conventions, such as biomedicine, chemistry, or the social sciences, remains to be verified, and may require domain-specific adaptations at the observation-extraction stage.

Backbone scale. We evaluate ObSchema with mid-scale backbones (14B–30B dense and MoE models, and GPT-5-nano). How schema quality scales with substantially larger models such as GPT-5 full or Qwen3-235B is an open question, particularly for the knowledge-intensive decisions in value-aware duplicate merging and aspect induction, where stronger parametric knowledge may further sharpen fine-grained structural calls.

Ethical Statement

All articles in our corpus are restricted to open-access publications under the Creative Commons Attribution (CC-BY) license, which permits redistribution and reuse with attribution. The corpus contains no personally identifiable information and no human-subject data. We release all DOIs of papers used for the study. The LLM-as-judge reliability study (Appendix E) involves annotators with graduate-level training in the relevant scientific domains. Annotators were informed of the research purpose and the rubrics they would apply, and their labels were used solely for evaluating the judging protocol. No demographic or otherwise identifying

information was collected. AI writing assistance was used only for language polishing.

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A Motivations of Corpus-Grounded Schema Induction	

This appendix further discusses why scientific entity attributes need to be organized as corpus-grounded schemas rather than treated as either generic domain knowledge or isolated local extractions. In this work, a schema is a type-level attribute organization for a target entity type. It defines the reusable attribute space in which entity-property observations from different papers can be normalized, compared, integrated, and queried. Such schemas are therefore useful for cross-document information integration, scientific text understanding, question answering, and downstream research workflows that require consistent access to entity-property knowledge.

Scientific attribute schemas are difficult to construct because the relevant properties of an entity type are broad, fine-grained, and corpus-dependent. In our setting, local paragraphs often report an attribute together with its value, unit, and measurement or experimental conditions. The desired schema, however, should not simply preserve every local mention as a schema node. It should capture recurring type-level attributes, separate attributes

from conditions, normalize heterogeneous surface forms, and organize the resulting attributes into coherent scientific dimensions. This motivates a corpus-grounded schema induction problem rather than a simple schema generation or extraction problem.

LLM-generated schemas can be too generic. A natural approach is to directly ask an LLM to list the attributes of a target entity type, for example the properties of a battery cell. Such a schema can reflect broadly known domain concepts, but it is not necessarily grounded in the papers being analyzed. The generated attributes may be too general, may miss properties that are salient in the target corpus, or may use a granularity that does not match how the corpus reports measurements and configurations. This is especially limiting for scientific corpora, where different collections may focus on different material systems, experimental protocols, performance criteria, or application scenarios. Therefore, relying only on an LLM’s parametric knowledge does not provide sufficient control over corpus-specific attribute coverage.

Passage-level open extraction can be too specific. Another approach is to extract attributes independently from each passage using an open-schema LLM extractor. This provides strong local coverage, but the resulting inventory is not directly usable as a global schema. Local extractions often preserve passage-specific wording and conditions, producing labels such as *capacity at 0.5C* or *capacity retention after 300 cycles*. These labels may be valid descriptions of local observations, but they mix attributes with measurement contexts and lead to fragmented schema nodes across documents. Conversely, lexically similar labels may still denote different scientific quantities, such as *specific capacity*, *areal capacity*, and *capacity retention*. Thus, passage-level extraction supplies useful observation evidence, but additional aggregation and refinement are needed to turn these observations into reusable type-level attributes.

These limitations motivate the observation-to-schema view adopted by ObSchema. Rather than using an LLM-generated attribute list as the final schema, accepting raw passage-level attributes as schema nodes, or requiring experts to enumerate the complete attribute space in advance, ObSchema treats local entity-property mentions as observation evidence. Local observations provide coverage of what the corpus actually studies; corpus-level evi-

dence determines which candidates become recurring type-level attributes; and schema refinement uses values, units, definitions, and conditions to merge duplicates and organize attributes into coherent scientific dimensions. In this way, the induced schema is neither purely generic nor overly local, but grounded in the corpus while remaining reusable across documents.

B Implementation Details

This appendix provides implementation details for our method, including the batch-wise pseudocode, the evidence score used for attribute promotion, and the prompt templates used in each module.

B.1 Pseudocode

Algorithm 1 summarizes the overall batch-wise schema induction procedure described in Section 3. Given document batches and a target entity type, our method first extracts entity-centric observations from each paragraph using the current schema as soft guidance. The extracted raw attributes are then consolidated into canonical candidates, classified into the evolving hierarchy, and stored in a pending pool until they accumulate sufficient evidence for promotion. After each batch, the schema is refined through conservative duplicate merging and adaptive grouping, which keeps the hierarchy compact while avoiding a fixed-depth structure.

B.2 Evidence Score

Definition. For a candidate attribute a assigned to category c , let d_a denote the number of distinct source documents that mention a , and let e_a denote the number of distinct entity mentions supporting a . Let M_c be the maximum entity support among current leaf attributes in category c .

We compute the promotion score from two complementary signals. The first signal measures whether the attribute recurs broadly across both documents and entity mentions:

$$\text{pop}(a) = \frac{\log(1 + \min(d_a, e_a))}{\log(1 + k_{\text{seed}})},$$

where k_{seed} is a corpus-scale normalization constant. In our experiments, we set k_{seed} to the batch size. Taking the minimum of d_a and e_a prevents an attribute from receiving a high score simply because it is repeatedly extracted from a single document or a single entity mention.

Algorithm 1: Overview of ObSchema

Input: Document batches $\{B_t\}_{t=1}^T$; target entity type τ ; coarse categories $\mathcal{C}$; initial schema $\mathcal{S}_0$; threshold θ

Output: Hierarchical entity-centric schema $\mathcal{S}_T$

Initialize pending pool $\mathcal{P}_0 \leftarrow \emptyset$ and observation store $\mathcal{O}_0 \leftarrow \emptyset$;

for $t = 1$ **to** T **do**

- Extract observations $(e, a_{\text{raw}}, v, c, Q)$ from paragraphs in B_t using $\mathcal{S}_{t-1}$ as soft guidance;
- Update observation store $\mathcal{O}_t$;
- Consolidate raw attribute names into canonical candidates;
- foreach** *candidate attribute* a **do**
 - Classify a into the current schema by category routing and top-down descent;
 - Add a to the pending pool if it is valid;
- foreach** *pending attribute* $a \in \mathcal{P}_{t-1}$ **do**
 - Update its support statistics and compute $s(a)$;
 - if** $s(a) \geq \theta$ **then**
 - Promote a into the schema;
- Refine the schema with conservative duplicate merging and adaptive grouping;

Perform final global cleanup on $\mathcal{S}_T$;

return $\mathcal{S}_T$;

The second signal measures the local prominence of the attribute within its category:

$$\text{dist}(a) = 1 - \exp(-\kappa e_a / M_c),$$

where κ controls the saturation rate. We use $\kappa = 15.0$ in all experiments. This term increases rapidly when an attribute first accumulates entity-level support, and then saturates as its support approaches the leading leaf attribute in the same category.

The final promotion score is the geometric mean of the two terms:

$$s(a) = \sqrt{\text{pop}(a) \cdot \text{dist}(a)}.$$

A pending attribute is promoted into the schema when $s(a) \geq \theta$. We do not tune θ and use $\theta = 0.15$ as the default throughout our experiments.

Hyperparameter Studies. We study the sensitivity of schema quality and schema size to the promotion threshold θ by sweeping $\theta \in \{0.10, 0.13, 0.15, 0.18, 0.20\}$ with all other components fixed. Figure 5 shows that quality and size respond very differently to θ . Quality remains largely flat across the swept range, with Uniqueness dropping only at the most permissive setting and Sibling Coherence declining monotonically as the gate tightens. Schema size, by contrast, decreases

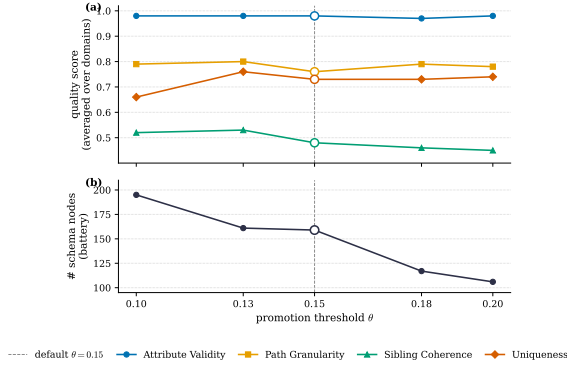


Figure 5: Effect of the promotion threshold θ on schema quality and size. (a) Quality metrics averaged over battery science and geoscience. (b) Number of promoted schema nodes on the battery domain, which decreases substantially across the same range.

Metric	minimal	low	medium
Attribute Validity	0.885	0.959	1.000
Path Granularity	0.789	0.857	0.833
Sibling Coherence	0.750	0.800	0.800
Uniqueness	0.628	0.721	0.750

Table 7: Effect of reasoning effort on schema quality, using GPT-5-nano on the Battery domain with all other components fixed. Gains are concentrated in the step from *minimal* to *low* and largely saturate thereafter.

substantially as θ increases. Reading the two panels together, the reduction in size at low θ co-occurs with a Uniqueness improvement, consistent with the gate suppressing redundant candidates; the further reduction at high θ co-occurs with the Sibling Coherence decline, suggesting the gate begins to remove evidence-supported attributes that contribute to within-category structure.

B.3 Reasoning Effort

In our main experiments we use the lowest reasoning effort on reasoning-capable backbones, for two reasons. First, several of our backbones are not reasoning models at all, so a low setting keeps the comparison across model families uniform. Second, information extraction is generally regarded as a task that is not reasoning-intensive (Sprague et al., 2025). To verify that this choice does not understate our results, we vary reasoning effort on GPT-5-nano over the Battery domain across three levels (*minimal*, *low*, *medium*) with all other components fixed.

As shown in Table 7, reasoning effort clearly helps: moving from *minimal* to *low* improves ev-

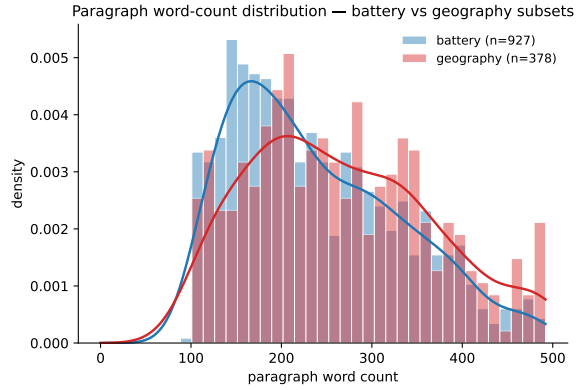


Figure 6: Density histograms with Gaussian-KDE overlays of paragraph word counts in the two final subsets: battery science (927, blue) and geography (n = 378, red). The x-axis is clipped at the 99.5th percentile.

ery metric substantially. The gains largely saturate beyond *low*, however. From *low* to *medium*, scores are stable or only marginally higher, and Path Granularity slightly decreases. This indicates that ObSchema does not require an unusually large reasoning budget to be effective, and is consistent with recent findings that test-time compute scaling yields limited returns on knowledge-intensive tasks (Zhao et al., 2026).

C Dataset Construction Details

Filtering criteria. To isolate entity-centric content, we retain passages describing entities of the target type with extractable scientific properties, measurements, configurations, or other entity-centric observations, and discard those that primarily contain background discussion, unrelated methodology, references, or generic narrative text.

Paragraph length distribution. Figure 6 plots the paragraph length distribution in each domain. The corpus exposes schema induction to a range of compression styles, from short, information-dense passages to longer paragraphs interleaving many properties.

D Baseline Details

Both TaxoAdapt and Chain-of-Layer were originally designed to operate on a pre-given concept set rather than on raw paragraph-level text. To apply them to our setting, we first run the base LLM on each paragraph in a single pass to extract candidate attribute mentions, and then feed the resulting attribute set into each baseline’s taxonomy construction procedure. All baselines are given the

same target entity type, coarse category set, corpus paragraphs, and output format. For methods not originally designed for scientific attribute schemas, we adapt their prompts to explicitly request measurable entity attributes and hierarchical organization. We also use the same backbone LLM and decoding setup.

Chain-of-Layer. Chain-of-Layer (Zeng et al., 2024) induces a taxonomy in a top-down, layer-by-layer fashion via an in-context-learning prompt that exposes the partial hierarchical structure to the LLM at each step.

TaxoAdapt. TaxoAdapt (Kargupta et al., 2025) dynamically adapts an LLM-generated taxonomy to a target corpus, expanding along multiple semantic dimensions through iterative hierarchical classification.

Iterative LLM Induction. This baseline directly leverages LLMs’ inherent capacity for structured knowledge organization (Yang et al., 2026): paragraphs are streamed one at a time to an LLM that maintains and updates a single running schema.

E LLM-as-Judge Protocol

E.1 Reliability Study

To assess the reliability of our LLM-as-a-judge evaluation protocol, we conduct two complementary studies. First, we randomly sample 100 judgment units for each metric, covering both battery science and geoscience, and ask annotators who have graduate degrees with relevant domain backgrounds to label them using the same rubrics as the LLM judge. Second, we repeat the judging process five times with GPT-5-mini under different random seeds and compute the standard deviation of each metric score. As shown in Table 8, the LLM judge achieves 78–85% agreement with human annotators on attribute validity, path granularity, and uniqueness, and 70% on sibling coherence, with mean standard deviations below 0.06 across all metrics. Attribute validity and uniqueness are the most stable, reflecting the relatively localized nature of these judgments (a single node, with or without one comparison). Sibling coherence shows lower agreement and higher variance, consistent with prior observations that comparing relative abstraction levels across multiple sibling nodes is inherently more subjective (Kargupta et al., 2025). We accordingly interpret sibling coherence differences within ± 5 points as inconclusive in the main

Metric	Agreement	Mean σ	Max σ
Attribute Validity	82%	0.019	0.032
Path Granularity	78%	0.037	0.050
Sibling Coherence	70%	0.053	0.136
Uniqueness	85%	0.032	0.064

Table 8: Reliability of the LLM-as-a-judge evaluation protocol. Agreement reports the agreement rate between human annotators and the LLM judge over $N = 100$ randomly sampled judgment units per metric. Mean σ and Max σ report the mean and maximum standard deviation of metric scores across five GPT-5-mini judging runs with five different random seeds.

results, and treat uniqueness as our most reliable structural metric.

E.2 Rubric Prompts

Below are the prompt templates used by our LLM-as-judge protocol. Each template is instantiated by substituting the listed input variables; the judge is asked to give a brief reason followed by a single tag drawn from a fixed scale.

Attribute validity

You are auditing a node in an attribute schema for entity type τ . Decide whether the node is a VALID attribute of τ -type entities.

A node is VALID iff its label names one of:

- an intrinsic property, characteristic, or performance metric of a τ -type entity that can take values in research papers;
- a composition or structural aspect of a τ -type entity;
- a component, part, or material of a τ -type entity.

A node is INVALID if it is any of:

- a measurement condition;
- too generic to be measurable.

Inputs: entity_type, parent_label, path, label, description.

Output: a brief reason (1–2 sentences), then one tag on its own line: <valid> or <invalid>.

Sibling coherence

You are auditing a set of sibling attribute nodes (children of a parent node) in an attribute schema for entity type τ . Decide the COHERENCE of these siblings: whether they are all at the same level of specificity, granularity, or abstraction under the parent.

Use a 4-level scale:

- <strong_sibling_coherence>: all siblings at the same level of specificity; they partition the parent’s space cleanly;
- <reasonable_sibling_coherence>: mostly coherent; one or two outliers in specificity;
- <weak_sibling_coherence>: considerable inconsis-

tency; several siblings out of place;
• <no_sibling_coherence>: highly inconsistent.
Inputs: entity_type, parent_label, parent_description, children_block.
Output: a brief reason, then one tag on its own line.

Path granularity

You are auditing a path in an attribute schema for entity type τ . This schema is not a strict is-a taxonomy; a child node may relate to its parent in several equally valid ways:

- a more specific KIND of the parent (is-a refinement);
- a PROPERTY or ASPECT of the parent;
- a COMPONENT or SUB-PART of the parent;
- a specific MEMBER of the group the parent names.

All of these count as valid hierarchical steps.

Decide whether the path has GOOD granularity: at each parent $\rightarrow$ child step, the child must stay WITHIN THE SCOPE of the parent—it narrows to a more specific sub-aspect of what the parent covers. The path has BAD granularity if any step:

- jumps to a concept not covered by the parent’s scope;
- merely renames the parent (the child is the same concept, not narrower);
- moves to a true sibling—a concept at the same level as the parent rather than contained inside it.

Inputs: entity_type, path_block (root $\rightarrow$ leaf, each step formatted as “label: description”).
Output: a brief reason, then one tag: <good_granularity> or <bad_granularity>.

Uniqueness

You are auditing whether a node in an attribute schema has a SEMANTIC EQUIVALENT elsewhere in the schema.

A node has an equivalent if there is another node in the schema (other than its own ancestors or descendants) that refers to the SAME underlying attribute concept, even if the label or wording differs. Parent-child relations are NOT considered equivalents.

Inputs: entity_type, label, description, path, schema_block (one node per line, “path: description”).
Output: a brief reason, then one tag: <has_equivalent> or <no_equivalent>.

F Case studies

F.1 Schema Case.

Figure 7 presents a partial schema induced for the battery domain. The schema separates measurable cell behaviors under cell_performance from structural and compositional aspects under cell_configuration. Within the performance branch, attributes such as specific_capacity, theoretical_capacity, and capacity_retention are organized under meaningful intermediate groups, while the configuration branch organizes components such

“The assembled LiFePO₄/Li cell showed a specific discharge capacity of 110 mA · h · g⁻¹ at 2 C over 200 cycles.”

ENTITY: LiFePO₄/Li cell

PATH: cell_performance ► energy_metrics ► specific_capacity

VALUE: 110 mA · h · g⁻¹

CONDITIONS: c_rate=2 C; cycle_number=over 200 cycles

“A practical-level full cell with a high-loading AC cathode provided an areal capacity of 2.8 mAh cm⁻² with 88% capacity retention after 300 cycles at 0.2 C.”

ENTITY: practical-level full cell with a high-loading AC cathode

PATH: cell_performance ► cycling_stability ► capacity_retention

VALUE: 88 %

CONDITIONS: cycle_number=after 300 cycles; c_rate=0.2 C

Table 9: Entity-property extractions from two representative battery-paper sentences.

as electrolyte, anode, and cathode. This illustrates that ObSchema induces a structured entity-centric representation that captures recurring scientific dimensions rather than a flat list of attribute names. Table 9 shows examples of observation extraction.

F.2 Generalization Beyond Scientific Domains

ObSchema is designed for scientific attribute schema induction, but we also probe how far it carries beyond that setting. We run it on a profile subset of MAVe (Yang et al., 2022), an attribute-value-extraction dataset built from Amazon product profiles, using Qwen3-30B-A3B and comparing against TaxoAdapt, our strongest baseline. All components and hyperparameters are unchanged.

As shown in Table 11, ObSchema outperforms TaxoAdapt on all four metrics. Absolute scores on the structural metrics are lower than in our scientific domains for both methods, which we attribute to the flatter and more surface-variable organization of product attributes. MAVe also provides a human-designed gold schema, allowing direct coverage evaluation. Across the five product types we examined, ObSchema recovers 100% of the gold attributes while surfacing additional valid ones. For guitar strings, beyond the six gold attributes it induces coating, core_shape, scale_length, tension, tuning, pack_size, and end_type, all valid on manual inspection. It also improves on the gold schema: the gold attributes type and acoustic/electric overlap, since “acoustic” is effectively a value of type, and ObSchema’s value-aware merging consolidates

Attribute	ObSchema	TaxoAdapt	Chain-of-Layer	Iterative LLM
<i>ionic conductivity</i>	configuration / ionic conductivity	performance / ionic conductivity	performance / ionic conductivity	performance / ionic conductivity
<i>voltage plateau</i>	not promoted	performance / discharge voltage / discharge voltage plateau	performance / voltage range / voltage plateau	not present
<i>cathode material</i>	configuration / cathode material	configuration / cathode material	configuration / cathode catalyst material	not present
<i>electrolyte composition</i>	configuration / electrolyte composition	configuration / electrolyte composition	configuration / electrolyte composition	not present

Table 10: Representative attribute placements across methods. ObSchema makes more semantically grounded category decisions, such as placing *ionic conductivity* under configuration, but can miss rare canonical attributes when they do not pass the evidence threshold.

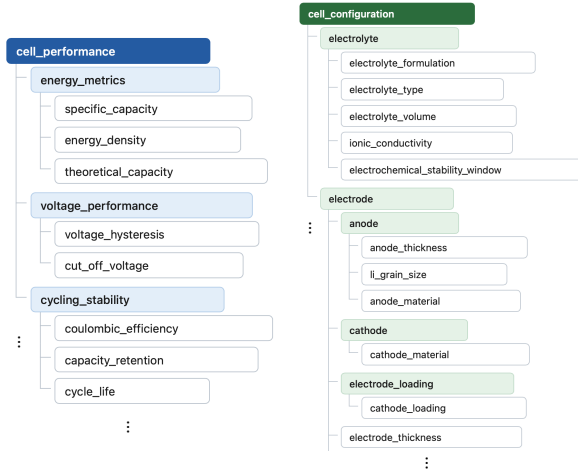


Figure 7: Partial view of the schema induced for the battery domain. Two top-level branches partition the schema by semantic role (cell performance (blue) for measurable behaviors and cell configuration (green) for structural composition)

Method	AV	PG	SC	Uniq.
TaxoAdapt	0.954	0.581	0.545	0.426
ObSchema	0.961	0.769	0.650	0.668

Table 11: Schema induction quality on a 5000 product profiles from MAVE subset with Qwen3-30B-A3B.

them into a single attribute, *instrument_type*, preserving the gold semantics with clearer naming.

F.3 Error Analysis

We conduct an attribute-level error analysis on the battery-domain schema induced by ObSchema. The analysis complements the automatic schema-quality metrics by examining three questions: (i) which reference attributes are missed by the final schema, (ii) whether promoted nodes contain invalid or redundant attributes, and (iii) how representative attributes are placed by ObSchema compared

Decision	#Attr.	%
Captured	42	70.0
Captured via merge	8	13.3
Pending only	7	11.7
Low-match ambiguous	2	3.3
Never extracted	1	1.7
Captured in final schema	50	83.3
True false negatives	10	16.7

Table 12: False-negative analysis against a 60-attribute battery reference list. Most reference attributes are captured by the final schema. The main source of misses is the evidence-based promotion filter: several real but rare attributes are extracted into the pending pool but do not accumulate enough support to be promoted.

with baseline schemas.

False negatives. We construct a reference list of 60 battery attributes curated by a domain expert with graduate-level training in battery science. Representative examples are presented in Table 10. The expert was given a sample of paragraphs from our corpus and asked to enumerate the attributes they consider essential for characterizing battery cells, using the annotation interface shown in Figure 8. The expert had no access to any ObSchema output, so the reference list is constructed independently of the induced schemas. The reference contains 32 cell-performance attributes and 28 cell-configuration attributes, each with a short description and surface-form synonyms. We match each reference attribute against both the final promoted schema and the full working schema, which retains pending nodes. Matching uses three signals: exact or normalized name matching against node names and aliases, embedding similarity between attribute descriptions, and a raw-attribute trace over extracted attributes.

Label	#Nodes	%
Core	69	84.1
Valid niche	12	14.6
Redundant	1	1.2
Invalid	0	0.0
Core + valid niche	81	98.8

Table 13: False-positive audit over promoted battery-schema nodes. ObSchema produces no invalid promoted attributes and only one residual duplicate, indicating that evidence-based promotion and schema refinement maintain high attribute validity while still retaining specialized scientific attributes.

Table 12 summarizes the results. ObSchema captures 50 of the 60 reference attributes in the final schema: 42 are directly matched to promoted nodes, and 8 are correctly absorbed through alias merging. Among the remaining 10 true false negatives, the dominant cause is the evidence-based promotion filter. Seven attributes are extracted and placed in the pending pool but do not receive enough corpus-level evidence to be promoted (e.g. *critical_current_density*, *nucleation_overpotential*, *self_discharge_rate*, *charge_overpotential*). Only one reference attribute is never extracted, and two are borderline cases whose nearest matches are semantically related but not identical. We find no cases where a reference attribute is extracted but discarded by the coarse-category gate. This suggests that the main coverage bottleneck is not semantic routing, but the conservative evidence threshold used to prevent rare or overly local attributes from entering the global schema.

False positives. We next audit all promoted attribute nodes in the ObSchema battery schema. Each node is independently classified into one of four labels: *core*, *valid niche*, *redundant*, or *invalid*. The judge is given the node name, parent, description, document count, representative values, and the full schema outline.

As shown in Table 13, 69 of 82 promoted nodes are core battery attributes, and 12 are valid niche attributes. These niche attributes are specialized but still scientifically meaningful, such as SEI composition attributes, Warburg diffusion impedance, coating-related properties, and chemistry-specific transport quantities. We find no invalid promoted nodes and only one redundant node, *ohmic drop*, which overlaps with existing resistance-related attributes. This audit supports the high attribute-

validity scores reported in the main experiments: the promoted schema is not simply large or permissive, but consists almost entirely of valid battery-cell attributes.

Attribute-level placement. We further compare how ObSchema and the baselines place representative battery attributes. Several core attributes, such as *capacity retention*, *cycle life*, *coulombic efficiency*, and *energy density*, are consistently placed under the performance branch by all methods, suggesting that highly frequent battery metrics form a stable method-invariant core.

The differences are more informative for attributes whose semantic role is less obvious from the surface name. For example, ObSchema places *ionic conductivity* under cell configuration, while the baselines place it under battery performance. This is more consistent with its role as a property of the electrolyte or material configuration rather than a direct cell-level performance output. In contrast, *voltage plateau* is a real miss for ObSchema: it appears in the working trace but does not pass the promotion threshold, showing that evidence-based filtering can suppress rare-but-canonical attributes. We also observe baseline-specific errors: Chain-of-Layer over-specializes *cathode material* into *cathode catalyst material*, while Iterative LLM Induction misses central configuration attributes such as *cathode material*, *anode material*, and *electrolyte composition*. These cases illustrate that ObSchema’s main errors are conservative coverage errors, whereas baselines more often suffer from misplaced, overly specific, or missing structural nodes.

Overall, the error analysis shows that ObSchema’s failures are primarily conservative coverage errors. Rare but valid attributes may remain pending when their corpus-level evidence is insufficient, as in the case of *voltage plateau*. In contrast, we observe low precision failures: the promoted schema contains no invalid nodes and only one residual duplicate. This trade-off is aligned with the goal of inducing a reusable type-level schema rather than exhaustively preserving every local property mention. A natural future extension is to soften the promotion rule for rare but canonical attributes, for example by incorporating domain-specific priors or an expert allowlist.

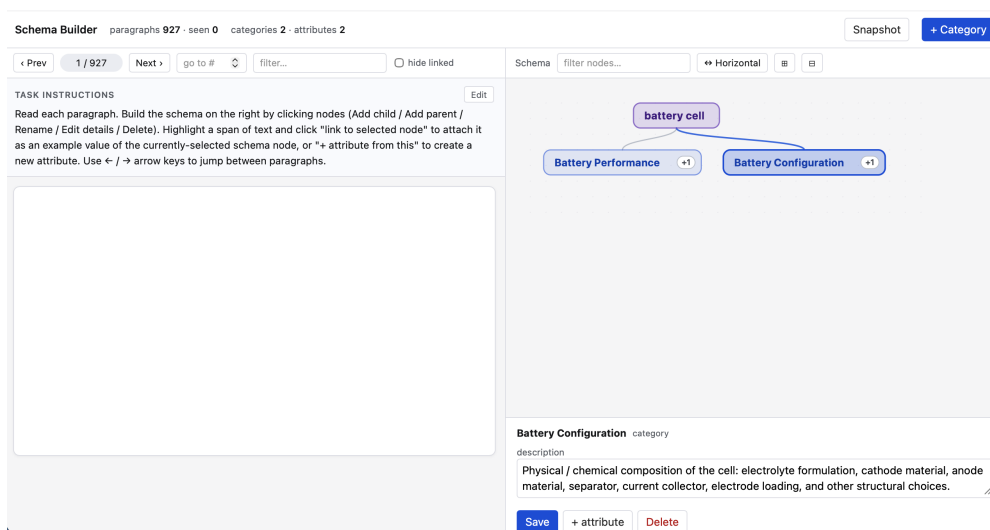


Figure 8: Annotation interface for constructing the entity-centric attribute schema.

G All Prompt Templates

This appendix section shows the main prompt templates used.

Observation Extraction

System. You are extracting entity-centric observation records from one paragraph of scientific text. The target entity type is {entity_type}. For each distinct target entity mentioned in the paragraph, extract the measured or reported properties of that entity, together with their values, the experimental or measurement conditions under which the values were obtained, and a definition of each attribute. Respond only with one valid JSON array.

Task.

Entity type: {entity_type}
Coarse categories: {categories_block}
Current schema, used as soft guidance: {schema_block}
Domain notes: {domain_notes}
Guidance examples for schema usage, attribute naming, and canonicalization: {schema_usage_examples}
Guidance examples for distinguishing attributes, values, and conditions: {attribute_condition_examples}
Paragraph: {paragraph}

Instructions. Extract only observations about the target entity type. Each observation should describe a property of the entity, not a process step, characterization method, background claim, or test protocol by itself. Use an existing canonical attribute name from the schema when the paragraph expresses a known attribute; otherwise, create a concise snake_case name for the new attribute. For every extracted attribute, also provide a one-sentence definition describing what the attribute measures or characterizes in the paragraph context.

Experimental or measurement settings should be attached as conditions rather than emitted as standalone attributes. If the paragraph contains no valid entity-property observation, return an empty array.

Output.

[json_format]

Category Gate

System. You are routing a candidate attribute through two checks. First, decide whether the candidate describes a property, state, or characteristic of the entity, rather than a process, method, procedure, measurement protocol, or characterization technique. Second, if it is a valid property, route it to one of the listed coarse categories. Respond only with one JSON object.

Task.

Entity type: {entity_type}
Candidate categories: {categories_block}
Candidate attribute: {x_brief}
Guidance examples: {category_gate_examples}
If the candidate is not a valid property, output:

```
{
  "category": "NONE",
  "reason": "<one short line>"
}
```

If it is a property, route it to the best category:

```
{
  "category": "<exact category name, or NONE>",
  "reason": "<one short line>"
}
```

Within-Category Descent

System. You are placing a new attribute into an existing scientific attribute schema. For each candidate child branch, you will see either a grouping node or a leaf attribute. Decide whether the new attribute should descend into one candidate branch or remain at the current level. Respond only with one JSON object.

Task.

Entity type: {entity_type}
Category: {category}
Current schema context: {current_block}
New attribute: {x_brief}
Candidate child branches: {candidates_block}
Guidance examples: {descent_examples}

Output.

```
{
  "reasoning": "<brief explanation>",
  "chosen": "<candidate_name or NONE>"
}
```

Schema-level duplicate merging

System. You are verifying whether embedding-retrieved attributes are true duplicates. The embedding model has already retrieved a small candidate group, but similarity is only used for recall. Your task is to merge attributes only when they denote the same underlying scientific quantity. Respond only with one JSON object.

Task.

Entity type: {entity_type}
Category: {category_name}
Candidate group: {cluster_block}
Each attribute includes its name, definition, representative values, units, and supporting observation count.

Guidance examples.

Examples of attributes that should be merged: {positive_merge_examples}
Examples of attributes that should not be merged: {negative_merge_examples}
Examples involving modifiers, normalization bases, subjects, measurement directions, temporal meanings, and unit differences: {modifier_and_unit_examples}

Output.

```
{
  "reasoning": "<brief comparison>",
  "merge_groups": [
    {
      "members": ["<name>", "<name>"],
      "keep_name": "<one member name>",
      "confidence": 0.0,
      "justification": "<why>"
    }
  ]
}
```

If no attributes should be merged, output:

```
{
  "reasoning": "...",
  "merge_groups": []
}
```

Organizing Aspect Discovery

System. You are identifying organizing aspects for a set of scientific attributes. Given a broad category and its current leaf attributes, identify a compact set of aspects that describe the main scientific dimensions covered by the attributes. These aspects are intermediate signals for grouping and are not necessarily inserted into the schema. Respond only with one JSON object.

Task.

Entity type: {entity_type}
Category: {category_name} —
{category_description}

Attributes currently under this category: {attrs_block}
Examples of good organizing aspects: {positive_aspect_examples}
Examples of bad or overly broad organizing aspects: {negative_aspect_examples}

Output.

```
{
  "aspects": [
    {
      "name": "<aspect name>",
      "description": "<scientific dimension>"
    }
  ]
}
```

Canonical Group Consolidation

System. You are consolidating candidate organizing-parent proposals into a compact set of optional grouping nodes. Merge near-synonymous proposals, reject incoherent ones, and keep only grouping nodes that capture a clear scientific aspect shared by multiple attributes. Respond only with one JSON object.

Task.

Current parent: {parent_name} —
{parent_description}
Candidate organizing-parent proposals: {proposals_block}
Examples of good grouping nodes: {positive_group_examples}
Examples of bad grouping nodes: {negative_group_examples}
Target number of grouping nodes, used only as soft guidance: {target_high}

Output.

```
{
  "canonical_groups": [
    {
      "name": "<canonical group name>",
      "description": "<what it groups>"
    }
  ]
}
```

Iterative LLM Induction

System. You are building a structured schema for scientific information extraction. Respond only with a valid JSON object. No extra text, no markdown fences.

Task. You are maintaining a schema for extracting structured measurement data from scientific papers about {entity_type}s. Update the schema based on the new document. Return the complete updated schema-.

Inputs.

Coarse categories (every attribute must be assigned to exactly one): {categories_block}
Current schema: {schema_json}
New document: {text}

Rules for attributes.

- Add new attributes drawn only from what the document actually measures or describes.
- Assign each attribute to one of the coarse categories above (use the category name verbatim in the category field). Do not invent new categories.

- If a new name is semantically equivalent to an existing one, fold it in (no duplicates). Prefer the more general / canonical form.
- Use short lowercase snake_case names. Do not bake test conditions into the attribute name.

Organize the attributes into a hierarchy —you decide the structure:

- Each attributes entry is a tree node with an optional children list.
- Nest a more specific attribute under a broader one whenever that captures real structure (a specialization, sub-aspect, or component of the parent).
- Introduce intermediate grouping attributes where several attributes share a broader concept that is worth naming.
- A top-level attribute (directly under a category) carries a category field; nested children do not.
- No depth limit and no requirement to keep it flat.

Conditions vs. qualifiers.

- role = "condition": external experimental parameter; changing it means measuring the *same* quantity under different circumstances. Conditions are reusable across attributes.
- role = "qualifier": redefines *what* quantity is measured; changing it yields a categorically different quantity. Bind each qualifier to its base attribute via `qualifier_of`.

For each attribute, list its typical condition-type names in `typical_conditions`. Echo the categories list back unchanged.

Output.

```
{
  "entity_type": "{entity_type}",
  "categories": [...],
  "attributes": [
    {
      "name": "<attr>", "category": "<category>",
      "typical_conditions": ["<cond>", ...],
      "children": [
        {
          "name": "<sub_attr>",
          "typical_conditions": [...],
          "children": [...]
        }
      ]
    }
  ],
  "condition_types": {
    "<name>": {
      "role": "condition" | "qualifier",
      "qualifier_of": "<attr> or null"
    }
  }
}
```

as a supporting mention or promote it into a new attribute.

H Schema Interface

Figure 8 shows the web interface used to construct the expert reference list. The left pane presents corpus paragraphs one at a time, while the right pane hosts an editable schema tree. Annotators read each paragraph and build the schema by adding, renaming, or deleting nodes, and can highlight a span of text to either link it to the selected node